

Weld Schedule Manager (Fill & Cap)

WPS No	July test weld 4	Position	Fixed 5G - Down	Diameter (in)	48	Wall (mm)	13.7
Process	GMAW	Specification	API1104	Filler Wire	Lincoln Pipeliner ER70-S6 (1.2mm)	Shield Gas	Ar/C02 85/15
Bevel Design	30 degree	Material Grade	X70	Trade Name	Sure-arc	Completed by	Ben Hatch



0° Position

Travel Speed 15

Torch Angle 0

30° Position

Travel Speed 16

Torch Angle -5

Wire Diameter

1.2

Direction of Travel

Vertical Down

Process

Pulse

Deposition Ratio

11

Shield Gas Flow (L/min)

26

60° Position

Travel Speed 18

Torch Angle -10

90° Position

Travel Speed 20

Torch Angle -10

120° Position

Travel Speed 18

Torch Angle -10

150° Position

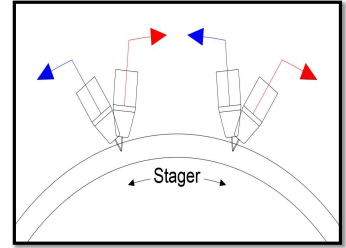
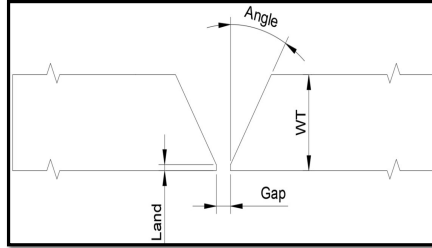
Travel Speed 14

Torch Angle -10

180° Position

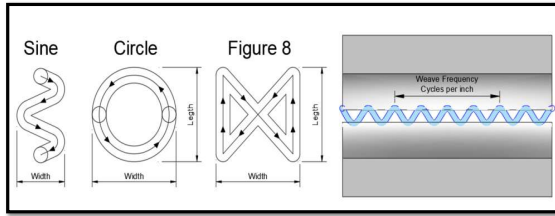
Travel Speed 10

Torch Angle -10



Joint Geometry		Tolerance (+ / -)	
Wall Thickness (mm)	13.7	1	0.5
Land (mm)	1	0.5	0.5
Root Gap (mm)	3	1	1
Bevel Angle (°)	30	5	5
Hi / Low (mm)		1.5	1.5

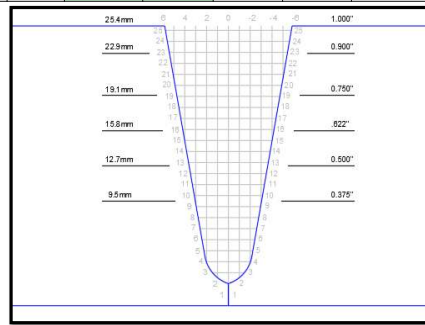
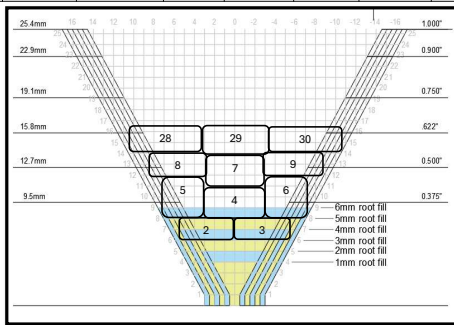
Stop / Start	
Stager the Stop-Start (mm)	30
Over run the next pass (mm)	12
Lead in at re-start (mm)	10



Weld Characteristics	
Deposition Efficacy	0.98
Pass Volume (mm2)	11.3
Min Cap Width (mm)	22
Root Depth (mm)	6
Total Passes (Inc root)	10

Total Arc Time (mins)	23.4
Wire Consumption (lb)	6
Wire Consumption (kg)	3

	Weave Adjustment					Weld Adjustment					Cleaning / Grinding					Comments
	Weave Pattern	Width (mm)	Legth (mm)	Dwell (sec)	Frequency / inch	Deposition Adjustment	Travel Speed Adjustment	Pass Volume	Pass Time	Arc Legth	Grind Prior to Weld Pass	Grind After Weld Pass	Travel Speed (in/min)	Weave Amplitude	Frequency Hz	
Hot Pass 0mm	Sine	2.0	2.0	0.0	12	-20%	-20%	9.8	2.9	2	OFF	ON	75	1	2	
Hot Pass 1mm	Sine	2.0	2.0	0.0	12	-10%	-10%	11.0	2.6	2	ON	OFF	75	1	2	
Hot Pass 2mm	Sine	2.0	2.0	0.0	12	-10%	-10%	11.0	2.6	2	ON	OFF	75	1	2	
Hot Pass 3mm	Sine	2.0	2.0	0.0	12	-10%	-10%	11.0	2.6	2	ON	OFF	75	1	2	
Hot Pass 4mm	Sine	2.0	2.0	0.0	12	0%	0%	12.2	2.3	2	ON	OFF	75	1	2	
Hot Pass 5mm	Sine	2.0	2.0	0.0	12	0%	0%	12.2	2.3	2	ON	OFF	75	1	2	
Fill Passes	Sine	1.5	1.5	0.0	12	0%	0%	12.2	2.3	3	ON	OFF	75	1	2	
Cap Passes	Sine	2.5	2.5	0.0	12	-10%	0%	11.0	2.3	4	ON	ON	75	1	2	



Pass #	On/Off	Pass Type	0mm		1mm		2mm		3mm		4mm		5mm		Comments
2	ON	Hot Pass	4.0	2.0	5.0	2.5	5.0	3.0	5.0	3.0	5.0	3.5	6.0	4.0	
3	ON	Hot Pass	4.0	-2.0	5.0	-2.5	5.0	-3.0	5.0	-3.0	5.0	-3.5	6.0	-4.0	
4	ON	Fill Pass	6.0	0.0	8.0	0.0	8.0	0.0	8.0	0.0	8.0	0.0	9.0	0.0	
5	ON	Fill Pass	6.0	3.5	8.0	4.0	8.0	4.5	8.0	4.5	8.0	5.0	9.0	5.5	
6	ON	Fill Pass	6.0	-3.5	8.0	-4.0	8.0	-4.5	8.0	-4.5	8.0	-5.0	9.0	-5.5	
7	ON	Fill Pass	8.0	0.0	9.0	0.0	9.0	0.0	9.0	0.0	9.0	0.0	10.0	0.0	
8	ON	Fill Pass	10.0	4.0	11.0	4.5	11.0	5.0	11.0	5.0	11.0	5.5	12.0	6.0	
9	ON	Fill Pass	12.0	-4.0	13.0	-4.5	13.0	-5.0	13.0	-5.0	13.0	-5.5	14.0	-6.0	
10	OFF	Fill Pass	12.0	0.0	13.0	0.0	13.0	0.0	13.0	0.0	13.0	0.0	14.0	0.0	
11	OFF														
12	OFF														
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24	OFF														
25	OFF														
26	OFF														
27.0	OFF														
28.0	ON	Cap Pass	14.0	6.0	14.0	6.5	14.0	7.0	14.0	7.0	14.0	7.5	14.0	8.0	
29.0	ON	Cap Pass	14.0	0.0	14.0	0.0	14.0	0.0	14.0	0.0	14.0	0.0	14.0	0.0	
30.0	ON	Cap Pass	14.0	-6.0	14.0	-6.5	14.0	-7.0	14.0	-7.0	14.0	-7.5	14.0	-8.0	

Notes

root fill and cap was visually acceptable

Tes was with ER70 S6 Pipeliner F&C to see if we could repeat test 3 that we did with surearc.
 We also removed fill 10 because it was over filled on weld July test 3, and we widened the cap by 1mm to try and flatten it.
 We reduced the deposition ratio to 11 to make up for the larger wire diameter.
 I reduced the over run from 20mm to 12mm to prevent the lump that was occurring